Fundamental Steps in Digital Image Processing

Rama Moorthy H

Dept of BCA, NIPE

Overview of Digital Image Processing

Two Broad Categories

- **Image-In/Image-Out:** Processes that take an image and output a modified image.
- **Image-In/Attributes-Out:** Processes that extract data or features from an image.

Note

Not all processes are applied in every case. The choice depends on the problem and application.

Fundamental Steps

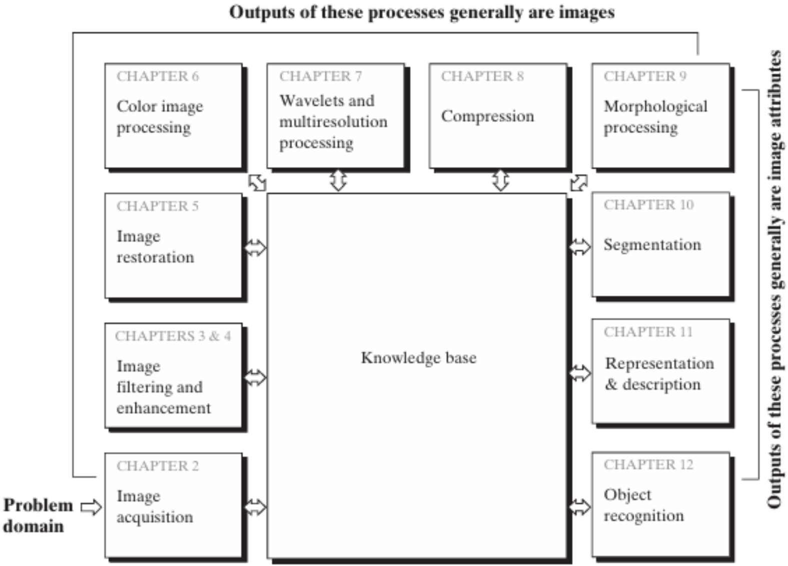

Image Acquisition

Definition

The process of capturing or receiving an image in digital form.

- Camera input
- Scanner
- Satellite feed

Preprocessing

May include operations like:

- Resizing
- Scaling
- Denoising

Image Enhancement

Goal

Improve image quality for a specific task.

Characteristics

- Problem-oriented
- Subjective (depends on human observer)

Examples

- Sharpening X-ray images
- Enhancing satellite infrared images

Image Restoration vs Enhancement

Image Enhancement

- Subjective
- Viewer decides image quality
- May involve contrast, brightness

Image Restoration

- Objective
- Based on mathematical models
- Removes known degradations (e.g., blur, noise)

Color Image Processing

Importance

- Widely used on the Internet and digital platforms.
- Color helps in identifying features of interest.

Key Topics

- Color models (RGB, HSV, CMY)
- Basic color operations

Wavelets and Multiresolution

Wavelet Use

- Compress image data
- Enable multi-level analysis

Applications

- Pyramid representations
- Image enhancement

Compression

Purpose

Reduce:

- Storage space
- Transmission bandwidth

Common Formats

- `.jpg` – JPEG standard
- `.png`, `.webp`, etc.

Morphological Processing

Purpose

- Analyze geometric structures
- Extract shape features

Segmentation

Definition

Partition an image into its constituent parts or objects.

Challenges

- One of the hardest tasks
- Accuracy impacts recognition success

Representation and Description

Representation

- **Boundary-based:** Edges, shapes
- **Region-based:** Texture, internal patterns

Description (Feature Extraction)

- Extracts quantitative attributes
- Used for classification and recognition

Object Recognition

Objective

Assign labels to segmented objects based on their features.

Examples

- "Tree"
- "Car"
- "Face"

Knowledge Base

Role of Knowledge Base

- Guides module operation
- Stores domain-specific data
- Controls module interactions

Summary

Key Takeaways

- Not all processes are needed in every case.
- Image viewing can occur after any stage.
- Complexity increases with task difficulty.